

Experiment No. : 10

Aim : To study a real-time application of Artificial Intelligence (AI) in the education sector through a published research paper and understand its architecture, working, benefits, and applications.

Topic :

“Artificial Intelligence in Education”

- A Review (Chen, Chen & Lin, IEEE Access, 2020)

Introduction:

Artificial Intelligence (AI) has transformed the education sector by improving administration, instruction, and learning. This review paper discusses how AI systems are applied in grading, adaptive learning, tutoring systems, and virtual classrooms. It emphasizes that AI enables personalization of content, automation of administrative tasks, and new teaching methods using intelligent systems, robots, and web-based platforms.

Literature Review:

- Early AI adoption in education began with **computer-aided instruction (CAI)** in the 1970s.
- AI evolved into **intelligent tutoring systems (ITS)** such as **AutoTutor**, **MATHia**, and **ActiveMath**.
- Web-based and adaptive learning platforms like **AIWBES (Adaptive and Intelligent Web-Based Educational Systems)** enabled personalized education.
- Recent advances include **chatbots**, **humanoid robots**, and **VR-based learning** that enhance teaching and student experiences.
- Studies show AI improves **efficiency**, **personalization**, and **global accessibility** of education.

Working of the System:

- 1) AI in Administration:
 - Automates grading, plagiarism detection (TurnItIn, Grammarly), and feedback.
 - Saves teachers' time and improves efficiency.
- 2) AI in Instruction:
 - Uses robots (cobots), chatbots, and simulation-based tools (VR/3D learning).
 - Enhances classroom teaching with personalized methods.
- 3) AI in Learning:
 - Adaptive systems like Knewton, Cerego customize lessons per student needs.
 - Web-based intelligent platforms allow access from anywhere.
 - Encourages deeper learning and student engagement.

Benefits of AI in Education:

- Efficiency: Reduces teacher workload in grading and administration.
- Personalization: Learners receive customized content based on ability.
- Global Access: Web-based systems eliminate geographical barriers.
- Engagement: VR and simulation create immersive learning experiences.
- Data-driven Insights: AI analyzes student performance for better outcomes.

Applications (Real-time):

- AI-powered grading tools (Grammarly, TurnItIn, Ecree).
- Intelligent tutoring systems (MATHia, AutoTutor, ActiveMath).
- Adaptive learning platforms (Knewton, Cerego, Immersive Reader).
- Robots in education (cobots for pronunciation and spelling practice).
- VR/3D simulations in medical and engineering education.
- Chatbots for student support in online learning environments.

Conclusion :

This case study highlights how AI is revolutionizing education by improving administration, teaching, and learning. Intelligent systems provide personalized instruction, automate repetitive tasks, and enhance student engagement through simulations and adaptive platforms. AI in education demonstrates the practical impact of technology in shaping future classrooms—making learning more efficient, accessible, and tailored to individual needs.